CRYSTALLOGRAPHY & X-RAY DIFFRACTION — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: March 31, 2026

Diagram-based questions on Crystallography & X-ray Diffraction

Question 1 10.0 points

Examine the diagram below illustrating the geometry of Bragg's law, where an incident X-ray beam strikes a crystal plane and the diffracted beam is observed. Identify the angle of incidence, the spacing between planes, and the conditions for constructive interference according to the equation $n\lambda = 2d\sin\theta$. Explain how each component is represented in the diagram.

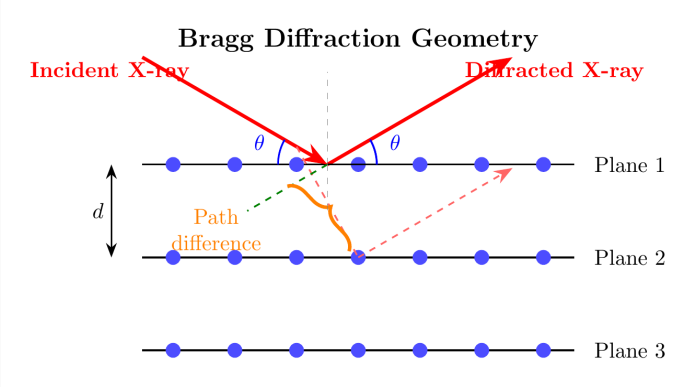

Fig. 1. Diagram illustrating the geometry of Bragg's law. Show an incident X-ray beam striking a crystal plane at an angle θ, with the diffracted beam also shown. Include multiple parallel crystal planes with spacing d between them. Label the angle of incidence θ, the plane spacing d, and the path difference for constructive interference. Indicate the condition for constructive interference using the equation nλ = 2dsinθ. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram incorrectly labels the angle of incidence with a. Regenerate using EXACTLY the values and sequences specified.

Question 2

10.0 points

Examine the diagram below of the reciprocal lattice and Ewald sphere construction. Determine where the reflections will occur when an X-ray beam is incident on a crystal. Define the conditions under which a specific point in the reciprocal lattice satisfies the Laue condition, and interpret its meaning in the context of X-ray diffraction.

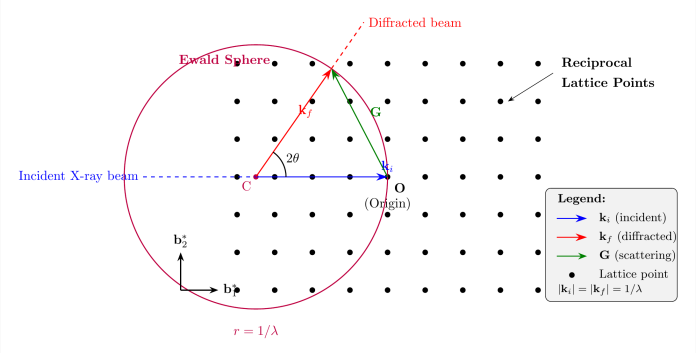

Fig. 2. A diagram showing the reciprocal lattice and Ewald sphere construction for X-ray diffraction. Include a lattice of points representing the reciprocal lattice, a circle representing the Ewald sphere, and an incident X-ray beam. Highlight the points where the Ewald sphere intersects the reciprocal lattice, indicating potential reflection points. Label the incident beam, reciprocal lattice points, and Ewald sphere for clarity.

Question 3 10.0 points

Interpret the diagram below that illustrates Miller indices (100), (110), and (111) planes in a cubic unit cell. Explain how these planes are oriented with respect to the cubic axes and how each set of indices determines the plane's orientation.

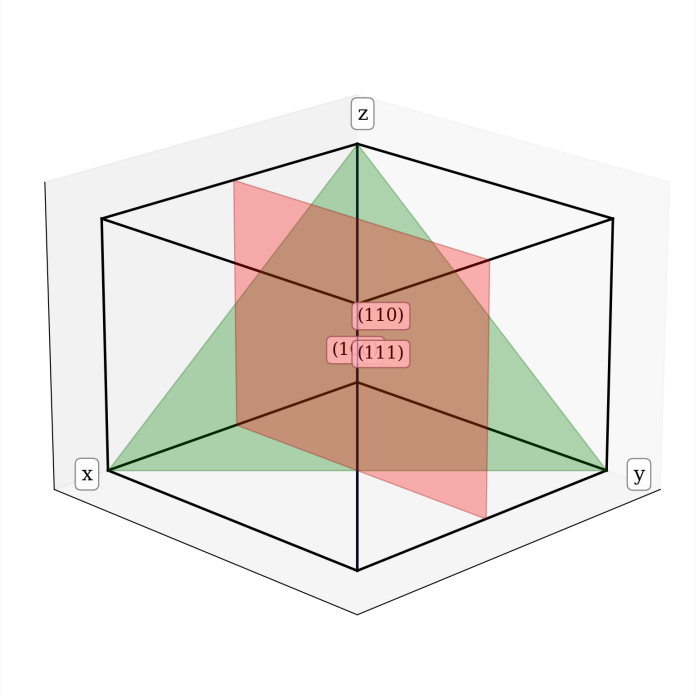

Fig. 3. A 3D diagram of a cubic unit cell showing the Miller indices (100), (110), and (111) planes. The cubic unit cell should be represented with its axes labeled. The (100) plane should be a flat plane cutting through the x-axis, the (110) plane should cut through both the x and y axes, and the (111) plane should cut through the x, y, and z axes. Each plane should be shaded or colored differently for clarity, and the orientation of each plane with respect to the cubic axes should be clearly visible. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram incorrectly represents the (10. Regenerate using EXACTLY the values and sequences specified. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram shows planes for (11. Regenerate using EXACTLY the values and sequences specified.

Question 4 10.0 points

Examine the diagram below of an FCC diffraction pattern. Identify the systematic absences in the diffraction peaks and explain the underlying cause of these absences based on the lattice plane reflections.

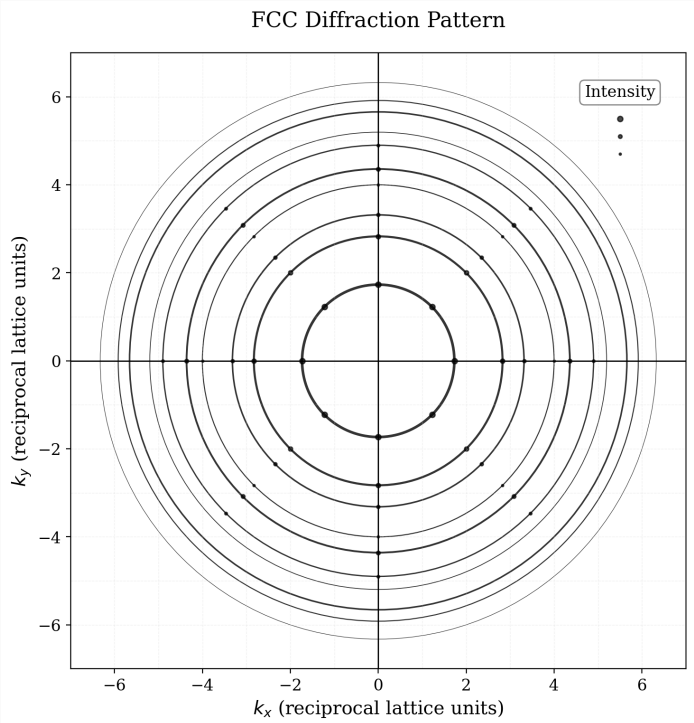

Fig. 4. Diagram of an FCC diffraction pattern showing diffraction peaks. The pattern should include labeled peaks corresponding to different lattice planes, with systematic absences indicated. This will help students identify and explain the absences based on lattice plane reflections. IMPORTANT: The diagram is missing required labels. The diagram is missing essential labels for. Ensure ALL structural components mentioned in the description are clearly labeled. IMPORTANT: The diagram is missing required labels. The diagram description requires labeled peaks corresponding to different lattice planes. Ensure ALL structural components mentioned in the description are clearly labeled.

Question 5 10.0 points

Examine the atomic arrangement of the BCC crystal in the diagram below. Use this to calculate the structure factor for a given set of Miller indices, assuming only atomic scattering factors are considered. Detail the steps involved in the calculation.

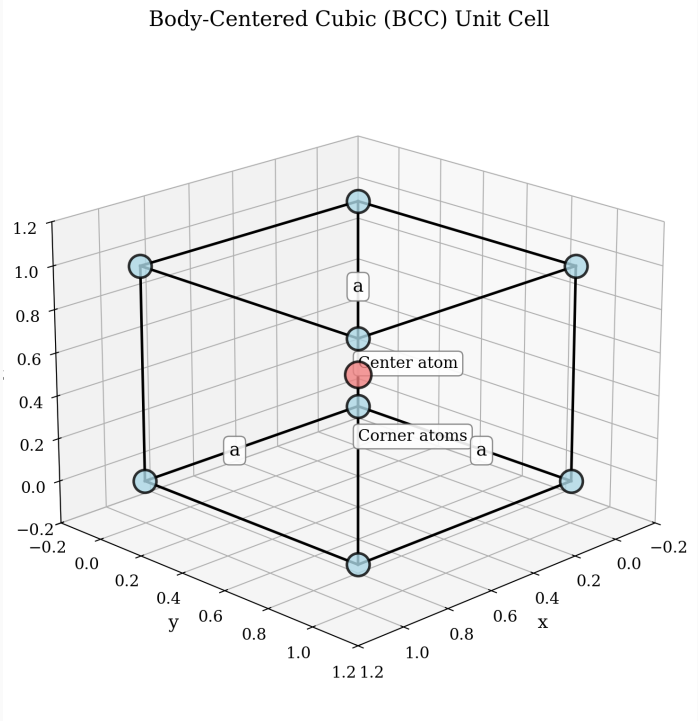

Fig. 5. 3D diagram of a Body-Centered Cubic (BCC) unit cell showing atoms at the corners and one atom at the center of the cube. Include labels for the lattice points and highlight the spatial arrangement to aid in understanding the calculation of the structure factor for given Miller indices. IMPORTANT: The diagram is missing required labels. The diagram description requires labels for. Ensure ALL structural components mentioned in the description are clearly labeled. IMPORTANT: The diagram is missing required labels. The diagram description requires labels for the lattice. Ensure ALL structural components mentioned in the description are clearly labeled.